Lab 7: Null Space and Inverses

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 21st, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Rank, Column Space, Null Space, and Inverses

- For an $n \times d$ matrix A, colsp$(A) \subseteq \mathbb{R}^n$ has dimension rank(A).
- nullsp$(A) = \{\vec{x} \in \mathbb{R}^d : A\vec{x} = \vec{0}\}$, and dim(nullsp$(A)$) $= d -$ rank(A).
- rank(A^T) = rank(A).
- A square matrix A is invertible if and only if it has full rank. Its inverse satisfies $AA^{-1} = A^{-1}A = I$.

Activity 1: PrairieLearn Practice Problems

We're testing out a new website for practicing linear algebra problems: PrairieLearn.

Click **this link** to access the relevant problems for this activity. It consists of 6 problems, each worth 1 point. The numbers in the problems are randomized; everyone will receive slightly different problems.

To get credit for Activity 1, you must **eventually** correctly answer all 6 problems, earning a score of 6/6. If you answer a problem incorrectly, click "New Variant" to generate a new version and then try again. There is no penalty for answering a problem incorrectly, as long as you eventually get it correct.

To be clear, you don't need to include anything in your PDF for Activity 1; we will manually verify that you've finished all 6 problems.

If you have trouble accessing PrairieLearn, message Suraj on Slack.

Activity 2: Null Space of a Matrix with Linearly Independent Columns

Let $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \\ 1 & 0 \end{bmatrix}$. What is nullsp$(A)$?

Activity 3: Null Space from Column Relationships

Let $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

This is the matrix from Lab 6. Recall that its first two columns form a basis for its column space and its rank is 2.

a) Fill in the blanks: nullsp(X) is a ____-dimensional subspace of ____.

b) Find a basis for nullsp(X). *Hint: You should be able to answer this without solving equations.*

Activity 4: The Systems of Equations View

Let A be an $n \times d$ matrix of rank r, and suppose there exist vectors $\vec{b} \in \mathbb{R}^n$ such that

$$A\vec{x} = \vec{b}$$

does not have a solution (meaning no $\vec{x}$ makes $A\vec{x} = \vec{b}$).

a) What are all inequalities ($<$ or $\leq$) that must be true between n, d, and r?

b) How do you know that $A^T\vec{y} = \vec{0}$ has solutions other than $\vec{y} = \vec{0}$?

Activity 5: Symbolic Inverses

Given that A is an invertible $n \times n$ matrix that satisfies $A^4 - 3A^2 + 2A - 4I = 0$, find an expression for A^{-1} in terms of A.

Activity 6: Basics of Invertibility

Suppose A is an $n \times n$ matrix. State as many of the equivalent conditions for invertibility as you can.

Activity 7: Invertibility of $X^T X$

a) Let X be an $n \times d$ matrix. Fill in the blanks: $X^T X$ is invertible if and only if X's columns are ____.

Activity 8: Doing Things Differently

Matrix multiplication does not always behave like multiplication of numbers. All matrices below are 2×2, and all vectors are in $\mathbb{R}^2$. Justify each answer. For a counterexample, give specific matrices or vectors and show the relevant products.

a) If $B\vec{v} = \vec{0}$, must $AB\vec{v} = \vec{0}$? If $AB\vec{v} = \vec{0}$, must $B\vec{v} = \vec{0}$? Address both directions.

b) Can $AB = 0_{2\times 2}$ even when neither A nor B is the zero matrix? Give an example or explain why it is impossible.

c) Expand $(A + B)^2$ using distributivity, keeping the factors in their original order. Is it always equal to $A^2 + 2AB + B^2$? If not, give a counterexample.